VBPM: Value-Based Personality Model

Abstract

Human behavior often diverges even when individuals face the same situation.
This divergence cannot be fully explained by differences in ability, information, or external rewards.
Instead, it frequently arises from differences in the structure of personal values—what people care about and how strongly they prioritize each value.

This study proposes the Value-Based Personality Model (VBPM), a minimal and computational framework that represents personality as a set of value dimensions and their associated weights.
Within this framework, action selection is formalized as the maximization of *internal consistency*: the degree to which an action aligns with an individual's value structure.
We define value criteria as pairs of value dimensions and weights, and compute internal consistency as a weighted sum over these criteria.

To demonstrate the model, we implement a simple “baby personality” with strong preferences for safety and comfort.
Given two scenarios, the model selects the action with the highest internal consistency, reproducing intuitive behavior in a minimal form.

VBPM is not a complete personality model; it is an initial, deliberately simplified step.
However, by isolating the core mechanism—value-based prioritization—it provides a clear and extensible foundation for future work in personality modeling, decision-making systems, and value-aligned artificial agents.

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1. Introduction

1.1 Background and Problem Setting

In organizations and social groups, it is not uncommon to observe situations in which highly capable individuals are marginalized or optimal proposals are rejected.
If people acted based on a shared set of values, such phenomena would be difficult to explain.

High performers tend to prioritize efficiency and outcomes, often presenting solutions that are objectively optimal.
When other members reject such proposals, the cause is not merely differences in ability or information-processing speed.
Rather, it is more natural to interpret these conflicts as arising from **differences in the prioritization of values**.

Importantly, this is not a matter of good or bad.
Some individuals prioritize change and innovation, while others value stability and order.
The latter often emphasize group harmony and the preservation of existing structures, and they choose actions consistent with those priorities.
From their perspective, such actions are entirely rational.

From this observation, the present study derives the following hypothesis:

> **Differences in human behavior arise from differences in values and their prioritization.**

People do not possess a single unified value; instead, multiple values coexist in parallel, and their relative importance varies across individuals.
Consequently, even when placed in the same situation, individuals with different value structures will choose different actions, sometimes resulting in friction or exclusion.

The aim of this study is to formalize this hypothesis as a **computational model that can be reproduced through mathematical formulation and implementation**.

1.2 Research Questions

This study seeks to answer the following questions:

- **Why do people choose different actions in the same situation?**
- **What internal structure gives rise to these differences?**
- **Can this structure be reproduced through mathematical formulation and implementation?**

1.3 Contributions

The main contributions of this study are as follows:

1. **We propose a new model—the Value-Based Personality Model (VBPM)—that formalizes values and their prioritization as a computational structure.**
2. **We show that action selection can be expressed as the maximization of internal consistency.**
3. **We demonstrate that this model is implementable and can be reproduced with minimal mathematical and computational components.**

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2. Core Hypothesis

2.1 Behavioral Differences Arise from Differences in Value Structures

Even when placed in the same situation, people often choose different actions.
Such differences cannot be fully explained by variations in ability or the amount of information available.
This study assumes that these behavioral differences arise from **differences in values and their prioritization**.

Individuals hold multiple values simultaneously and assign different levels of importance to each.
Values such as efficiency, stability, harmony, challenge, safety, and creativity all influence human behavior, but the degree to which each value is prioritized varies significantly across individuals.

Because the *structure of values*—the set of values and their relative priorities—differs from person to person, people choose different actions even under identical circumstances.
In other words, behavioral differences arise not from differences in values themselves,
but from differences in the **weights assigned to those values**.

2.2 Action Selection as the Maximization of Internal Consistency

When choosing an action, people do not rely solely on external rewards.
They tend to select actions that do not conflict with their own value structure.
This study refers to this tendency as **internal consistency**.

Internal consistency represents
"the degree to which an action aligns with the values an individual prioritizes."

From this perspective, action selection can be understood as follows:

> **People choose the action that best aligns with their own value priorities.**

This view of action selection as the maximization of internal consistency
naturally explains behaviors that cannot be accounted for by external reward maximization alone—
for example, choosing inefficient actions to maintain order,
or rejecting optimal solutions for the sake of group harmony.

2.3 Central Hypothesis of This Study

Based on the above, this study presents the following central hypothesis:

> **Differences in human behavior arise from differences in values and their prioritization,
> and action selection can be understood as a process that maximizes internal consistency with one's value structure.**

In the next chapter, this hypothesis is **formalized as a computational model**,
and we demonstrate that it can be **reproduced through implementation**.

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3. Model

3.1 Purpose of the Model

In the previous chapter, we proposed the hypothesis that differences in human behavior arise from differences in values and their prioritization.
In this chapter, we formalize this hypothesis as a **computationally tractable structure**.
To do so, we decompose values into *value dimensions* and *weights*, and define their combination as a unit called a **Value Criterion**.

3.2 Value Dimensions

A value dimension is defined as an
"independent axis for evaluating actions or situations."

Examples include:

- efficiency
- stability
- harmony

- safety
- challenge
- creativity
- fairness

These dimensions are independent of one another, and individuals evaluate actions by referencing multiple dimensions simultaneously.

In this study, each value dimension is represented as a function:

$$V_i(a) : \text{evaluation of action } a \text{ with respect to value dimension } i$$

3.3 Weights

Weights represent
"how strongly an individual prioritizes each value dimension."

Even if two individuals share the same value dimensions,
differences in their weights can lead to different behavioral choices.

Weights are represented as a vector:

$$w = (w_1, w_2, \dots, w_n)$$

Weights may change through experience,
and such changes manifest externally as changes in personality.

3.4 Value Criteria

The central concept of this study, the **Value Criterion**, is defined as follows:

> **Value Criterion = (Value Dimension, Weight)**
> That is,
> **an evaluative module composed of a value dimension and its associated weight.**

Formally:

$$C_i = (V_i, w_i)$$

Individuals possess multiple value criteria,
and they integrate these criteria when evaluating actions.

3.5 Internal Consistency

The internal consistency of an action a is computed by aggregating the evaluations of all value criteria, weighted by their importance:

$$Consistency(a) = \sum_i w_i \cdot V_i(a)$$

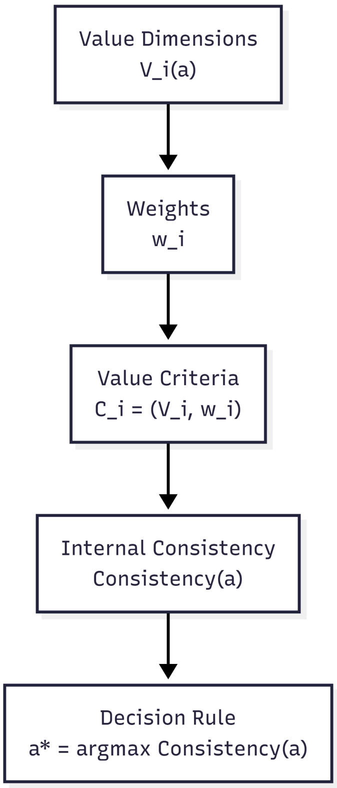
This represents
"the degree to which an action aligns with the values an individual prioritizes."

3.6 Decision Rule

Action selection is defined as the process of choosing the action that maximizes internal consistency:

$$a^\star = \arg\max_a Consistency(a)$$

This represents not the maximization of external rewards,
but the maximization of **alignment with one's internal value structure**.



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4. Implementation

4.1 Implementation Principles

The model proposed in this study decomposes values into *value dimensions* and *weights*,
and formulates action selection as the maximization of internal consistency (Chapter 3).

For implementation, we adopt the following assumptions:

- **The evaluation values $V_i(a)$ for each value dimension are provided externally by the scenario.**
  (e.g., a situation may be described as "safety = 0.9," "novelty = 0.2," etc.)
- **Individual differences exist only in the structure of the weights w_i.**
- **Value Criteria = (Value Dimension, Weight)** belong to the individual.
- **Scenarios provide only the outputs of value dimensions.**

These assumptions allow the implementation to remain fully consistent with the model while maintaining a minimal and transparent structure.

4.2 Data Structures

The implementation uses two primary data structures:

Personality

- A vector of weights assigned to each value dimension.
- Example: `{"safety": 0.9, "novelty": 0.3, ...}`

Scenario

- A mapping of value dimensions to their evaluation values $V_i(a)$.
- Example: `{"safety": 0.2, "novelty": 0.8}`

Both are represented using Python `dataclass` objects and dictionaries.

4.3 Computing Internal Consistency

As defined in Chapter 3:

$$Consistency(a) = \sum_i w_i \cdot V_i(a)$$

The implementation follows the algorithm below.

Algorithm 1: InternalConsistency(personality, scenario)

```
Input:
    personality.values : Dict[value_name, weight]
    scenario.scores    : Dict[value_name, score]

Output:
    consistency : float

Procedure:
    total ← 0
    For each value_name v in personality.values:
        w ← personality.values[v]
        s ← scenario.scores.get(v, 0)   # missing dimensions = 0
        total ← total + w * s
    Return total
```

4.4 Decision Rule

As defined in Chapter 3:

$$a^\star = \arg\max_a Consistency(a)$$

In implementation, we compute internal consistency for each scenario
and select the one with the highest score.

Algorithm 2: ChooseBestScenario(personality, scenarios)

```
Input:
    personality : Personality
    scenarios   : List[Scenario]

Output:
    best_scenario : Scenario

Procedure:
    scored_list ← empty list
    For each scenario in scenarios:
        score ← InternalConsistency(personality, scenario)
        Append (scenario, score) to scored_list

    Sort scored_list by score in descending order
    Return scored_list[0].scenario
```

4.5 Demonstration

In `prototype/baby_personality.py`,
we implemented a simple example representing a "baby personality."

- **Weights:** strong emphasis on safety and comfort
- **Scenarios:**
  - reaching for a colorful toy
  - staying close to the parent

After computing internal consistency for each scenario,
the model selected **"stay with parent."**

This result is consistent with a personality structure
that places high weight on safety.

4.6 Position of the Implementation

The algorithms presented in this chapter implement the model from Chapter 3
in its minimal functional form.

- Value Dimension → scenario evaluation values
- Weight → personality parameters
- Value Criterion → implicit pairing of the above

- Internal Consistency → weighted sum
- Decision Rule → argmax

The full implementation is provided in `prototype/baby_personality.py`.

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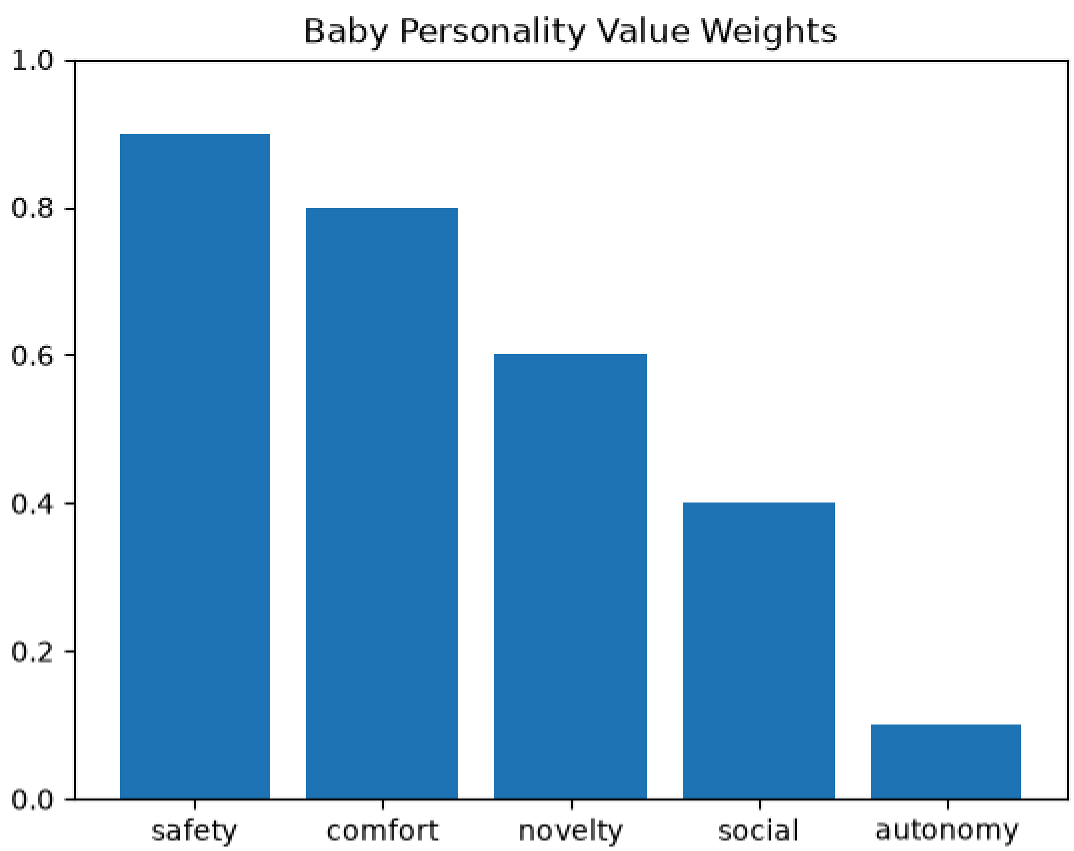
5. Demonstration

5.1 Experimental Setup

To demonstrate that the proposed model (VBPM) can reproduce actual patterns of action selection, we conducted a simple experiment using a “baby personality” as an illustrative example.

The baby personality is assumed to have the following value priorities:

- **safety**: 0.9
- **comfort**: 0.8
- **novelty**: 0.6
- **social**: 0.4
- **autonomy**: 0.1



This configuration reflects the intuitive assumption that babies strongly prioritize safety and comfort,
refer somewhat to novelty and sociality,
and place very little emphasis on autonomy.

We then presented two scenarios:

- **reach_for_toy**
  - reaching for a colorful toy
  - novelty: 0.8, safety: –0.1
- **stay_with_parent**
  - staying close to the parent
  - safety: 0.9, social: 0.4

Each scenario provides evaluation values $V_i(a)$ for the relevant value dimensions.

5.2 Internal Consistency Results

Based on the proposed model, we computed the internal consistency of each scenario.

reach_for_toy

$$Consistency = 0.9(-0.1) + 0.8(0) + 0.6(0.8) + 0.4(0) + 0.1(0) = 0.390$$

stay_with_parent

$$Consistency = 0.9(0.9) + 0.8(0) + 0.6(0) + 0.4(0.4) + 0.1(0) = 0.970$$

5.3 Action Selection

The comparison of internal consistency values is as follows:

- **reach_for_toy → 0.390**
- **stay_with_parent → 0.970**

Therefore, the model selects:

> **Selected action: stay_with_parent**

5.4 Interpretation

This result is consistent with the assumed value structure of the baby personality.

- Reaching for the toy scores high on novelty but low on safety.
- Staying close to the parent scores high on safety and sociality.
- Because the baby places the highest weight on safety, the model selects the latter.

Thus, the demonstration shows that:

> Given the specified value priorities and scenario evaluations, the model selects the action that is most consistent with those parameters.

This reflects the internal logic of the model rather than an attempt to reproduce actual human behavior.

5.5 Summary

This demonstration shows that the proposed model can reproduce
value-based action selection using only the minimal components of
“value dimensions × weights.”

In the next chapter, we discuss the implications and limitations of the model.

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6. Discussion

The Value-Based Personality Model (VBPM) proposed in this study aims to reproduce value-based action selection using minimal mathematical structure and implementation.
By representing personality as a combination of *value dimensions* and *weights*, and by formalizing action selection as the maximization of internal consistency, the model demonstrates that intuitive human behavior can be explained in a concise and computationally tractable manner.

However, this study represents only **a first step**, and several important limitations remain.

6.1 Exogeneity of Scenario Evaluations $V_i(a)$

In this study, the evaluation values $V_i(a)$ for each value dimension are treated as exogenous inputs provided by the scenario.

This is a reasonable simplification for implementation, but it differs substantially from actual human cognition:

- People do not evaluate situations objectively.
- Perception, attention, emotion, and memory all intervene.
- Values themselves can influence how situations are perceived.

Thus, treating $V_i(a)$ as fixed values constitutes a significant simplification of human decision-making processes.

6.2 The Unresolved Problem of Objectifying $V_i(a)$

If there existed a method to evaluate scenarios objectively,
VBPM could serve as a **highly consistent decision-making model**.

However, at present,
the assignment of $V_i(a)$ depends on the user’s subjective judgment,
which limits the reproducibility of the model.

Addressing this issue requires connecting VBPM to larger frameworks such as
world models or perception models that can generate evaluations from raw sensory or contextual information.

6.3 Interaction Between Values and Perception

Psychologically, values and perception are not independent; they interact.

- Individuals who prioritize safety tend to overestimate danger.
- Those who value novelty tend to perceive stimuli as more attractive.

In this study, we separated values (weights) from the evaluation function $V_i(a)$,
but this separation is a simplification intended to preserve the purity of the model.

Future extensions may incorporate models in which values influence perception itself.

6.4 Significance as a "Toy Model"

The implementation presented in this study is extremely minimal—simple enough to be called a *toy model*.

However, this simplicity provides several advantages:

- It enables intuitive understanding of how value structures influence behavior.
- It is applicable to education, games, and AI character design.
- It allows researchers and developers to freely extend and experiment with the model.

VBPM is not a complete personality model; rather, it serves as
a starting point for exploring value-based personality modeling.

6.5 Broader Implications for Social Friction and Decision-Making

Because VBPM formalizes behavioral divergence as a consequence of differences in value structures, improvements in the estimation of $V_i(a)$ or in the learning of value weights may allow the model to anticipate value-based conflicts or provide alignment-oriented recommendations. A more refined version of VBPM could therefore contribute—at least conceptually—to reducing interpersonal friction or supporting decision-making in multi-agent settings.

This remains speculative, as the present study does not empirically evaluate such effects. Nevertheless, the model's structure suggests potential relevance beyond individual action selection.

6.6 Future Directions

Future research may pursue the following directions:

- Generating $V_i(a)$ from perception models
- Modeling interactions between values and perception
- Extending the model to social interactions (multi-agent settings)
- Integrating weight learning into the core model
- Empirical validation using real behavioral data

These extensions may allow VBPM to evolve into a more realistic and powerful personality model.

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7. Conclusion (English Version)

This study proposed the Value-Based Personality Model (VBPM),
a framework for modeling action selection based on values and their prioritization.

VBPM decomposes values into *value dimensions* and *weights*,
and formalizes action selection as the maximization of internal consistency.
Through this formulation, the model captures the relationship between personality structure and behavior using minimal mathematical components.

The contributions of this study can be summarized as follows:

1. **We formalized the structure of values in a computationally tractable manner.**
2. **We demonstrated that action selection can be explained as the maximization of internal consistency.**
3. **We showed that the model can be implemented and reproduced with minimal code and mathematical structure.**

In the demonstration, we used a "baby personality" as an example to illustrate
how differences in value priorities influence action selection.
The results showed that VBPM naturally reproduces intuitive behavior.

However, this study represents only **a first step**,
and VBPM is not a complete personality model.
In particular, the question of how to assign evaluation values $V_i(a)$
is the most significant limitation of the model.
In the current implementation, scenario evaluations are provided externally,
and complex cognitive processes such as perception, attention, and emotion are not modeled.

Nevertheless, VBPM extracts the core structure of personality—
values × prioritization—
and provides a **minimal foundation** for computationally handling this structure.
As a "toy model," it offers a simple yet powerful starting point
for exploring value-based personality simulation.

Future work may include:

- Objectifying the evaluation function $V_i(a)$
- Modeling interactions between values and perception
- Extending the model to multi-agent social interactions
- Integrating weight-learning mechanisms
- Empirical validation using real behavioral data

These extensions may allow VBPM to evolve into a more realistic and powerful personality model.

This study presents a simple yet compelling perspective:
personality can be understood as a *structure of values*.
VBPM has the potential to serve as a foundation for value-based behavior modeling
across diverse domains, including personality research, AI character design,
and decision-making systems.
